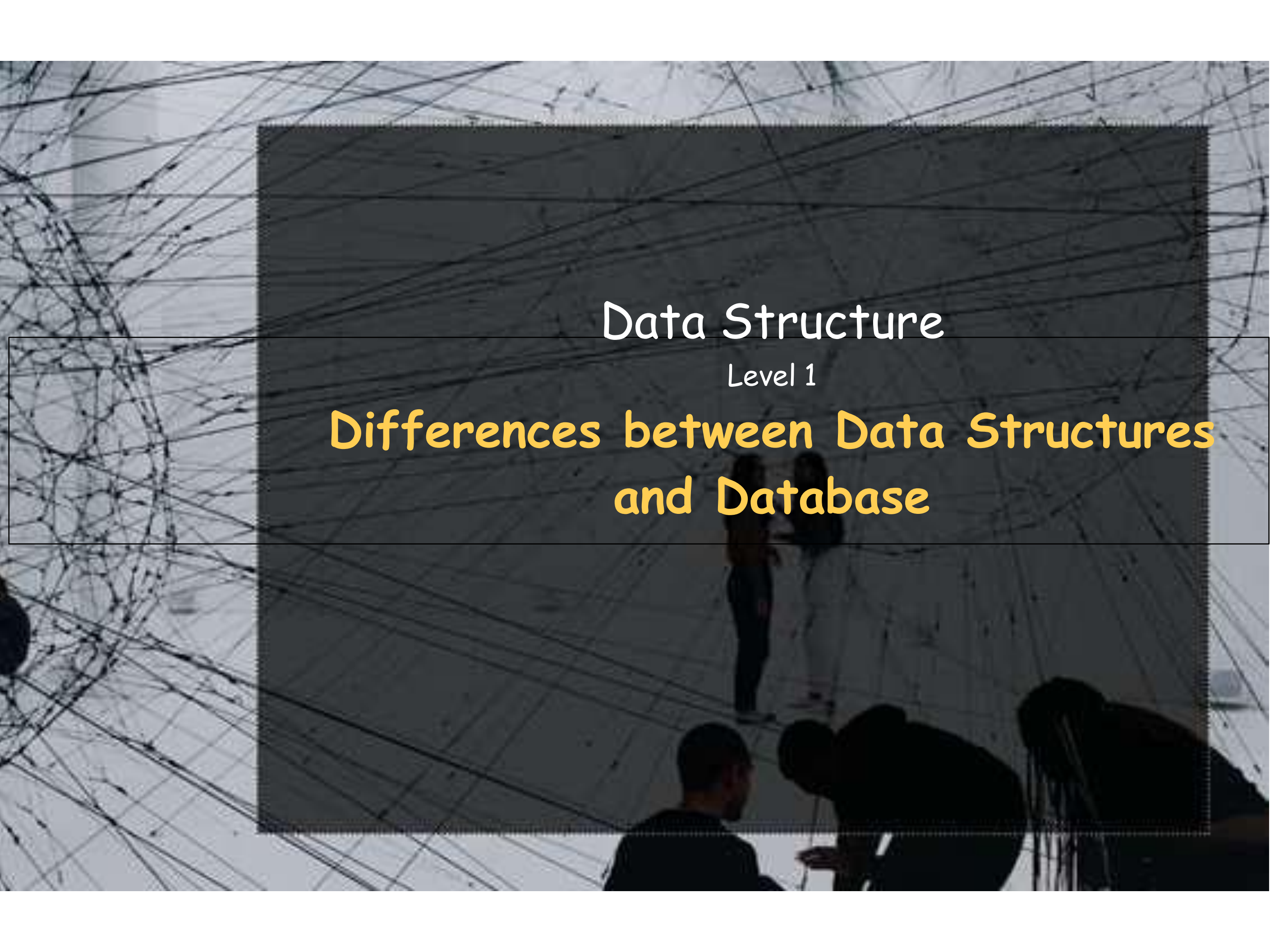


Data Structure Level 1

Differences between Data Structures and Database

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing the entire frame, resembling a web or a data structure. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark, semi-transparent rectangular box is centered on the slide, serving as a backdrop for the text.

Data Structure

Level 1

Differences between Data Structures and Database

Definition: Data Structure

- ◆ In computer programming, a **Data Structure** is a way of organizing and storing data in memory so that accessing and modifying data becomes efficient.

Explanation:

- ◆ Data structures are mainly used **inside programs** and exist during program execution.

Common Data Structures

-  Array
-  Linked List
-  Stack
-  Queue
-  Heap
-  Binary Tree
-  Graph

Example:

Using a **Stack** to manage function calls in a program.

Note:

Data structures focus on **memory efficiency and speed**.

Definition: Database

- ◆ A **Database** is a collection of data stored in an organized manner, usually in tables consisting of rows and columns, and managed using a **Database Management System (DBMS)**.

Explanation:

- ◆ Databases are used to store **large amounts of persistent data** that remain available even after the program ends.

Functions of DBMS

-  Define data
-  Store data
-  Modify data
-  Manipulate data
-  Manage access to data

Examples of DBMS

-  MySQL
-  Oracle Database
-  PostgreSQL

Example:

Storing user accounts and passwords in a *MySQL* database



Comparison

■ Data Structure

- ✓ Used within a program
- ✓ Stored in main memory (RAM)
- ✓ Temporary (exists during program execution)
- ✓ Focuses on performance and efficiency
- ✓ Examples: Array, Stack, Tree

■ Database

- ✓ Used for long-term data storage
- ✓ Stored on secondary storage (Disk)
- ✓ Persistent (data remains saved)
- ✓ Managed using DBMS software
- ✓ Examples: MySQL, Oracle

Interpretation

- ◆ Data structures help programs process data quickly, while databases help systems store and manage large volumes of data safely.

Warning

- ◆ A database **does not replace** data structures.
- ◆ Programs still use data structures internally even when working with databases.

Summary

- ✓ Data structures organize data **inside programs**
- ✓ Databases store data **outside programs**
- ✓ Data structures optimize speed and memory usage
- ✓ Databases ensure persistence, security, and scalability
- ✓ Both are essential and complement each other

Thank you 😊